# Empirical priors for high-dimensional problems

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# Introduction

- High-dimensional problems are everywhere now.
- Hopeless without some structural assumptions.
- *Sparsity* is a common type of structure, i.e., most signals are zero/small but a few are large.
- An assumed structure is "prior information," so Bayesians incorporate this structure via a prior.
- But except for this vague notion of structure, we usually don't know anything.
- And since it's high-dim, the choice of prior on the structure-specific parameters actually matters!

## Intro, cont.

- Priors that lead to posteriors with
  - relatively easy computation tend to be thin-tailed and have sub-optimal theoretical properties
  - good theoretical properties are heavier-tailed and tend to be more difficult to compute with.
- *Do we have to choose between theory and computation?*
- I say *NO!*
- Take-away messages:
  - *Tails don't matter if centering is right*.
  - To get the centering right, we need to use the data, i.e., an *empirical prior*.

# This lecture

- Background on Bayesian stuff[1]
- Sparse normal mean problem
  - empirical prior construction
  - posterior concentration results
  - uncertainty quantification
- Beyond the sparse normal mean problem
  - a piecewise constant structure
  - high-dim regression
- A general theory?
- Monotone density estimation
- Concluding remarks

[1] My understanding is that you've talked about Bayesian large-scale testing, local fdr, etc. I worked on such things before and, if you're interested, check out a recent review paper I wrote: `arXiv:1812.02149`

# Background

- Basic Bayesian approach:
  - statistical model with unknown $\theta \in \Theta$
  - get prior for $\theta$
  - combine prior with data/likelihood using Bayes's formula
  - use posterior distribution to make inference.
- Selling points:
  - conceptually easy to do
  - gives answers to basically any question in one stroke.
- But there are serious issues, so I'm not a Bayesian![2]
- The approach I'm going to talk about isn't Bayesian.
- But it's better than Bayesian and the Bayesians don't like it, so it's a win–win for me!

[2] https://www.researchers.one/article/2019-02-1

# Background, cont.

- Statistical model: $Y^n \sim \mathrm{P}_\theta^n$, joint density $p_\theta^n$.
- $\theta$ could be high- or infinite-dimensional.
- Let $\Pi_n$ be a prior distribution on $\Theta$.
- $L_n(\theta) = p_\theta^n(Y^n)$ is the likelihood.
- Bayes's formula defines the posterior distribution:

$$\Pi^n(d\theta) \propto L_n(\theta)\,\Pi_n(d\theta).$$

- Answers to statistical questions based on the posterior:
  - Estimators via posterior expectations
  - confidence sets using posterior quantiles
  - tests using posterior probabilities
  - predict a new $\tilde{Y}$ by integrating $p_\theta(\tilde{y})$ wrt posterior

# Background, cont.

- For low-dim problems, this is relatively easy:
  - prior distribution for low-dim $\theta$ doesn't matter too much
  - MCMC computation is fast
  - posterior has nice asymptotic properties almost automatically.
- In high-dim problems, however:
  - prior matters
  - computation is hard(er)
  - asymptotics are hard(er)
- "High-dim" means more parameters than data points.
- I discuss an important example below.

# Sparse normal means problem

- $Y^n = (Y_1, \ldots, Y_n)$, independent, $Y_i \sim \mathsf{N}(\theta_i, 1)$.
- High-dim, since each $Y_i$ has its own $\theta_i$.
- Assume vector $\theta$ is *sparse*, i.e., most $\theta_i$'s are 0.
- Non-Bayesian approach might do some kind of soft or hard thresholding, e.g., lasso, etc.
- How about a Bayesian?
- Express $\theta$ vector as a pair $(S, \theta_S)$:
  - $S \subseteq \{1, 2, \ldots, n\}$ denotes the location of non-zeros
  - $\theta_S$ the $|S|$-vector of non-zero values.
- Need a prior for $(S, \theta_S)$...[3]

[3] This discrete-$S$ approach is not the only Bayesian strategy, some adopt a "continuous shrinkage prior" like horseshoe, but I don't really like this.

# Sparse normal means, cont.

- Can use sparsity assumption to motivate a marginal prior for $S$, more on this below.
- But what about the conditional prior for $\theta_S$?
- Since the model is normal, computational simplicity suggests a conjugate normal prior for $\theta_S$.
- However, there are results[4] that say a prior for $\theta_S$ with thin tails leads to sub-optimal posterior concentration.
- Consequently, good theory requires tails to be heavier than normal, which makes computation difficult.
- Apparently theory and computation are at odds.
- *Or maybe not...*

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[4] e.g., Castillo & van der Vaart, *Ann. Statist.*, 2012

# Sparse normal means, cont.

- *Idea:* If sufficient information about $\theta_S$ is lacking to justify a prior, then why not let data help guide the choice?
- Marginal prior $\pi_n(S)$ for $S$:
  - $|S| \sim f(s) \propto n^{-as}$, $s = 0, \ldots, n$, where $a > 0$.
  - Given the size, $S$ is uniform over all configs of that size.
- Conditional prior for $\theta_S$, given $S$:

$$\pi_n(\theta_S \mid S) = \mathsf{N}_{|S|}(Y_S, \gamma^{-1} I_{|S|}), \quad \gamma \in (0, 1).$$

- Conditional prior for $\theta_S$, given $S$, times the marginal prior for $S$ gives an *empirical prior* for $\theta$ — call it $\Pi_n$.
- Intuition:
  - informative prior on the thing we know about, $S$;
  - "non-informative" on the thing we don't know about, $\theta_S$

# Sparse normal means, cont.

- Combine prior and likelihood in *almost* the usual way:

$$\Pi^n(d\theta) \propto L_n(\theta)^\alpha \, \Pi_n(d\theta), \quad \alpha \in (0,1).$$

- The power $\alpha$ is unusual, a regularizer, more on this later.
- Relatively simple computations thanks to:
  - closed-form expression for posterior of $S$,

$$\pi^n(S) \propto \pi_n(S) \, e^{-\alpha \|Y_{S^c}\|^2/2} (1 + \gamma^{-1}\alpha)^{-|S|/2}.$$

  - "closed-form" marginal posterior for each $\theta_i$, more later.
  - R code is on my website.
- Good theoretical properties too...

## Sparse normal means, cont.

- Summary of theoretical results spread over several papers.[5]
- *Recovery.*
  - If true $\theta^\star$ is $s^\star$-sparse, then $\Pi^n$ concentrates around $\theta^\star$ at the minimax rate $s^\star \log(n/s^\star)$, i.e., for some $M > 0$,

$$\sup_{\theta^\star : |S_{\theta^\star}| = s^\star} \mathsf{E}_{\theta^\star} \Pi^n(\{\theta : \|\theta - \theta^\star\|^2 > M s^\star \log(n/s^\star)\}) \to 0.$$

  - Prior doesn't know $s^\star$, so concentration rate is *adaptive*.
  - Posterior mean is an asymptotically minimax estimator.
  - $\Pi^n$ supported on a set of dimension roughly $s^\star$.

[5] arXiv:1304.7366, 1406.7718, 1604.05734, 1812.02150.

# Sparse normal means, cont.

Average value of $\|\hat{\theta} - \theta^\star\|^2$ in 100 replicates at $n = 200$.

| $s^\star$ | $10 = 0.05n$ | | $20 = 0.10n$ | | $40 = 0.20n$ | |
|---|---|---|---|---|---|---|
| $\theta^\star_{S^\star}$ | 7 | 8 | 7 | 8 | 7 | 8 |
| BL | 65 | 69 | 100 | 103 | 133 | 137 |
| DL | 16 | 14 | 33 | 31 | 66 | 60 |
| HS | 16 | 15 | 36 | 34 | 72 | 70 |
| *EB* | **13** | **12** | **24** | **24** | **49** | **49** |

(BL = Bayesian lasso; DL = Dirichlet–Laplace; HS = horseshoe)

# Sparse normal means, cont.

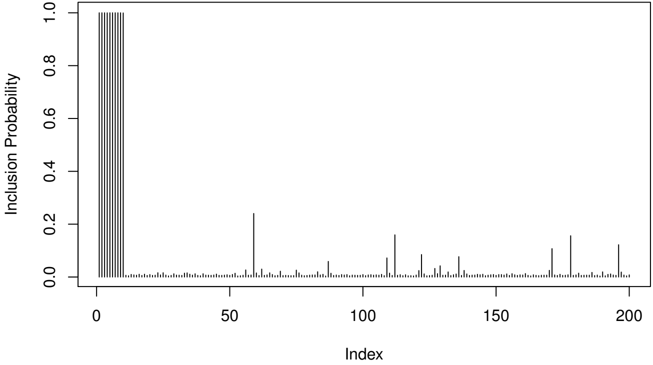
- *Structure Learning.*
  - Inclusion probability satisfies

$$p_i^n := \pi^n(S \ni i) \to \begin{cases} 0 & \text{if } \theta_i^\star = 0 \\ 1 & \text{if } |\theta_i^\star|^2 \gtrsim \log n. \end{cases}$$

  - If all non-zero $|\theta_i^\star|$ are "large enough," i.e., if

$$\min_{j \in S^\star} |\theta_j^\star| \gtrsim (\log n)^{1/2}, \qquad \text{("beta-min")}$$

  then $\pi^n(S = S^\star) \to 1$, where $S^\star = S_{\theta^\star}$.




$n = 200$ and $\theta_1^\star = \cdots = \theta_{10}^\star = 7$, others zero.

# Sparse normal means, cont.

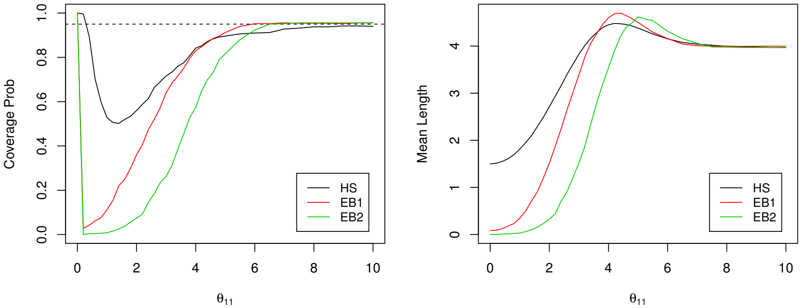
- *Uncertainty quantification.*
  - Marginal posterior for $\theta_i$ is available in "closed-form,"

    $$(1 - p_i^n)\,\delta_0 + p_i^n\,\mathsf{N}(Y_i, v_\alpha),$$

    where $p_i^n$ is the inclusion prob and $v_\alpha = (\alpha + \gamma)^{-1}$.
  - Note: if $\alpha \approx 1$ and $\gamma \approx 0$, then $v_\alpha \approx \mathsf{var}(Y_i)$.
  - Credible interval for $\theta_i$ fails to hit the target coverage prob asymptotically only if $\theta_i^\star \neq 0$ and $p_i^n \not\to 1$.
  - Conditions to avoid this, hopefully weaker than beta-min?
  - Specific results presented in `arXiv:1812.02150`, joint work with former NCSU student, Dr. Bo Ning.

# Sparse normal means, cont.

- $n = 200$, $\theta^\star =$ (five 7's, five 1.5's, $\theta^\star_{11}$, rest equal 0).
- Check coverage and length of 95% credible interval for $\theta_{11}$.
  - HS: $\sigma^2 = 1$ fixed, $\tau$ via MMLE, `horseshoe` package
  - EB1: $\alpha = 0.99$, $\gamma = 0.01$, from `arXiv:1304.7366`
  - EB2: $\alpha = 0.99$, $\gamma = 0.01$, based on `arXiv:1406.7718`



# Remarks about $\alpha$

- The power $\alpha$ might be a concern for some.
- Can absorb $\alpha$ into the empirical prior, regularization,

$$\Pi^n(d\theta) \propto L_n(\theta) \, \frac{\Pi_n(d\theta)}{L_n(\theta)^{1-\alpha}}.$$

- Anyway, the above convergence theory holds for any $\alpha \in (0,1)$ so there is no practical effect.
- $\alpha$ is only required for adaptive rates.
- It's also possible to do the necessary regularization in the prior for $S$, but that's more complicated.
- And there are situations where $\alpha < 1$ actually helps.[6]

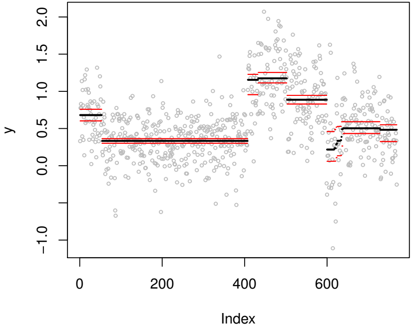
[6] e.g., Peter Grünwald's *SafeBayes*.

# Beyond sparse normal means

- The above formulation seems especially tailored to the sparse normal means problem.
- Can other problems/structures be handled similarly?
- The approach is actually quite general — more later.
- For now, two cases similar to that above:
  - normal mean with piecewise constant structure
  - high-dim linear regression

# Piecewise constant normal mean

- Same normal mean model but, instead of sparsity, assume that there are blocks of means having the same value.
- This model is for change-point problems, regression trees, copy number variation, etc.



# Piecewise constant, cont.

- Makes sense to write mean vector $\theta$ as $(B, \theta_B)$, where
  - $B$ is block configuration and
  - $\theta_B$ are the block-specific parameters.
- Sparsity is still relevant — not too many blocks.
- My empirical Bayes approach[7] proceeds as follows:
  - informative prior on $|B|$, number of blocks
  - uniform prior on block configuration, $B$, given the number
  - non-informative empirical prior on block-specific means, $\theta_B$
  - power $\alpha \in (0, 1)$ on likelihood.
- R code is on my website.

[7] arXiv:1712.03848

# Piecewise constant, cont.

- Let $\theta^\star$ denote the true piecewise constant mean vector, and $B_{\theta^\star}$ its "block configuration."
- If $\theta^\star$ is such that $|B_{\theta^\star}| > 2$, then the minimax rate is

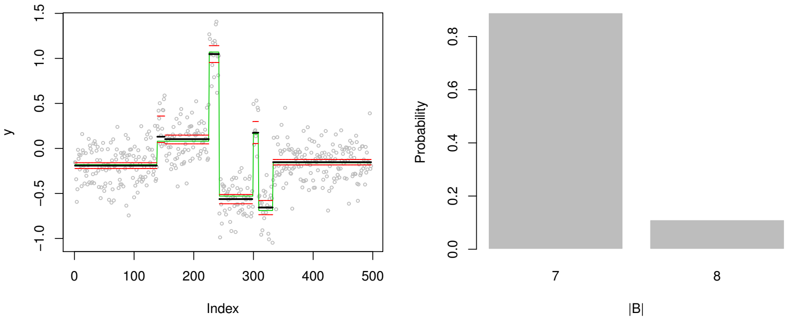
$$\varepsilon_n(\theta^\star) = |B_{\theta^\star}| \log\bigl(en/|B_{\theta^\star}|\bigr).$$

- Our theorem says the empirical Bayes posterior adaptively attains this rate, i.e.,

$$\sup_{\theta^\star} \mathsf{E}_{\theta^\star} \Pi^n(\{\theta : \|\theta - \theta\|^2 > M\varepsilon_n(\theta^\star)\}) \to 0, \quad n \to \infty.$$

- A consequence is that the empirical Bayes posterior mean, $\hat{\theta}_n$, is an adaptive minimax estimator.

- Standard test example:
  - $n = 497$ means; shown in green
  - $|B_{\theta^\star}| = 7$ blocks of unique means.
- Left plot shows $\hat{\theta}_n$ (black), 95% credible intervals (red)
- Right plot shows posterior distribution for $|B|$.



# High-dim linear regression

- Not surprisingly, a similar analysis can be carried out for the linear regression setting, i.e.,

$$Y = X\beta + \varepsilon,$$

where $Y$ is a $n$-vector, $X$ is a $n \times p$ matrix, with $p \gg n$.

- If $\beta$ is sparse, it makes sense to express $\beta$ as $(S, \beta_S)$.
- Empirical prior $\Pi_n$ for $\beta$:
  - Similar prior encodes sparsity in $S$,
  - $(\beta_S \mid S) \sim \mathsf{N}_{|S|}(\hat{\beta}_S, \gamma^{-1}(X_S^\top X_S)^{-1})$.
- Posterior: $\Pi^n(d\beta) \propto L_n(\beta)^\alpha \, \Pi_n(d\beta)$.
- As before, *is the posterior any good?*

# High-dim linear regression, cont.

- Good computational and theoretical properties.[8]
- Computation:
  - Conjugate normal prior, can integrate out $\beta_S$
  - Marginal posterior for $S$, simple Metropolis.
  - R code on my website.
- Convergence theory:
  - Minimax optimal rate with respect to $\|X\beta - X\beta^\star\|$.
  - Optimal rate for $\|\beta - \beta^\star\|$.
  - Model selection consistency.
- Other things (with NCSU students):
  - `arXiv:1810.00739` is on empirical correlation-adaptative priors with Mr. Chang Liu and Ms. Yue Yang
  - `arXiv:1903.00961` is on prediction with Ms. Yiqi Tang.

[8] `arXiv:1406.7718`

# High-dim linear regression, cont.

- Model selection results.[9]
- Rows of $X$: normal, pairwise correlation 0.25.
- $n = 200$, $p = 1000$, $\sigma^2 = 1$.
- True signal: $\beta^\star_{S^\star} = (0.6, 1.2, 1.8, 2.4, 3.0)^\top$.

| Method | $\mathrm{P}(\hat{S} = S^\star)$ | $\mathrm{P}(\hat{S} \supseteq S^\star)$ |
|---|---|---|
| BASAD | 0.930 | 0.950 |
| BASAD.BIC | 0.720 | 0.990 |
| BCR.Joint | 0.090 | 0.250 |
| Lasso.BIC | 0.020 | 1.000 |
| EN.BIC | 0.325 | 1.000 |
| SCAD.BIC | 0.650 | 1.000 |
| *EB* | 0.945 | 0.990 |

[9] Setting and results taken from Narisetty and He, *Ann. Statist.*, 2014.

# A general formulation/theory?

- Two examples above are still "normal."
- *Question:* Is it possible to do the empirical prior construction for more general kinds of problems?
- *Yes!* New strategy[10] based on an empirical KL condition can handle a lot, even nonparametric problems.
- This is messy, so my goal is a modest one:
  - share with you the basic idea
  - focus on intuition
  - one example: monotone density estimation[11]

[10] `arXiv:1604.05734`
[11] `arXiv:1706.08567`

# General theory

- In a Bayesian setting, good asymptotic properties require that the prior put sufficient mass near the true $\theta^\star$.
- Don't know $\theta^\star$ so it's not obvious how to arrange this.
- This brings us back to the tails of the prior...
- Remember:
  - *tails don't matter if the centering is right*.
  - The "right" centering depends on data.

# General theory, cont.

- The KL condition is standard in the literature.
- That is, if the target rate is $\varepsilon_n$, then the prior should assign sufficiently large mass to the set

$$\{\theta \in \Theta_n : K(p^n_{\theta^\star}, p^n_\theta) \leq n\varepsilon_n^2\}.$$

- If we knew $\theta^\star$, then just put all the mass there.
- But we don't know $\theta^\star$, hence tails need to be chosen carefully.
- Or we could try an informative center...
- Idea: Replace KL condition with an *empirical version?*

# General theory, cont.

- For a suitable sieve, $\Theta_n$, let be $\hat{\theta}_n$ a sieve MLE, and

$$\mathcal{L}_n = \{\theta \in \Theta_n : L_n(\theta) \geq e^{-cn\varepsilon_n^2} L_n(\hat{\theta}_n)\}.$$

- Requiring the prior to put sufficient mass on $\mathcal{L}_n$ amounts to an empirical KL condition.
- More specifically, I'll require the prior $\Pi_n$ satisfy

$$\Pi_n(\mathcal{L}_n) > e^{-Cn\varepsilon_n^2}, \quad \text{all large } n.$$

- Key consequence:
  - Let $D_n = \int_{\Theta_n} \{L_n(\theta)/L_n(\theta^\star)\} \, \Pi_n(d\theta)$.
  - If $n$ is large, then $D_n > e^{-(c-C)n\varepsilon_n^2}$.
  - Proof: $D_n > \int_{\mathcal{L}_n} \cdots \, \Pi_n(d\theta)$

# General theory, cont.

- Lower-bounding the posterior denominator $D_n$ is key.
- Comes basically for free under empirical KL condition.
- Roughly, the general result[12] assumes:
  - sieve has good approximation properties
  - empirical KL condition holds
  - prior tails aren't too heavy
- *If these conditions are satisfied for a particular sequence $\varepsilon_n$, then the posterior concentrates around $\theta^\star$ at that rate.*
- This is for a non-adaptive rate result; adaptation requires some more complicated conditions, omitted here.

[12] `arXiv:1604.05734`

# What do empirical priors look like?

- Empirical priors are "non-informative" but not in the traditional sense.
- That is, depending on the context, it may be that $\Pi_n$ is collapsing on $\hat{\theta}_n$ as $n \to \infty$, i.e., it's not diffuse.
- For example, in finite-dim problems, above procedure would suggest a prior with variance $O(n^{-1})$.
- Seems counter-intuitive, but:
  - No benefit to informative centering if the spread is wide!
  - Ultimately, rate isn't affected by prior concentration.
- How to choose constants, etc, for good coverage??

# Monotone density estimation

- Estimate a monotone non-increasing density $f$ with bounded support $[0, T]$; paper[13] handles more general case.
- Such densities have a mixture-of-uniforms representation

$$f(x) = \int k(x \mid \mu)\,\omega(d\mu), \quad k(x \mid \mu) = \mu^{-1}1(x \leq \mu).$$

- Take a sieve of finite mixtures

$$f_\theta(x) = \sum_{j=1}^{J_n} \omega_j k(x \mid \mu_j), \quad \theta = (\omega, \mu).$$

- Prior on $\theta$ leads to prior on $f$.

[13] `arXiv:1706.08567`

## Monotone density, cont.

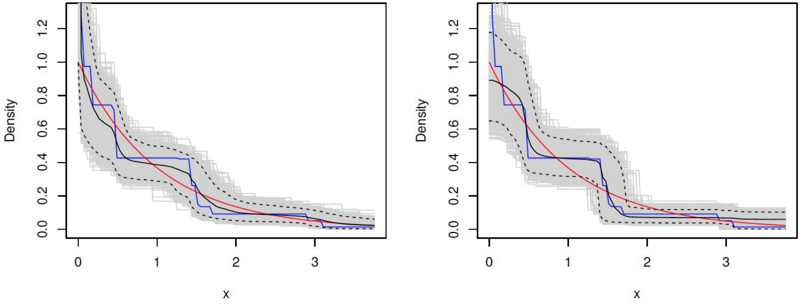
- Sieve: mixtures with $S$-many components, $\mu$'s in $[t, T]$.
- Let $\hat{\theta} = (\hat{\omega}, \hat{\mu})$ be sieve MLE.
- Prior for $\theta = (\omega, \mu)$:
  - $\omega$ and $\mu$ independent;
  - $\omega$ is $\mathsf{Dir}(\hat{\alpha})$ with $\hat{\alpha}_j = 1 + c\hat{\omega}_j$;
  - $\mu_j$ is $\mathsf{Par}(\hat{\mu}_j, \delta)$.
- Simple model, Pareto is conjugate to uniform kernel.
- Existing Bayes results *can't* use conjugate Pareto prior.
- Need to choose $(J, t, T, c, \delta)$ carefully, theory helps.
- R code on my website.

# Monotone density, cont.

- Target rate: $\varepsilon_n = (\log n)^{1/3} n^{-1/3}$, near minimax.
- Rate doesn't depend on true $f^\star$, no adaptation needed.
- Conditions:
  - $J_n \propto \varepsilon_n^{-1}$;
  - $c_n \propto n\varepsilon_n^{-2}$;
  - $\delta_n \log(T_n/t_n) \propto \log n$.
- Then there exists $M > 0$ such that

$$\mathsf{E}_{f^\star}\Pi^n(\{f : H(f^\star, f) > M\varepsilon_n\}) \to 0,$$

for any $f^\star$ monotone, supported on $[0, T^\star]$, with $f^\star(0) < \infty$.




(a) Empirical Bayes

(b) Dirichlet process mixture

Figure 1: Exponential example. Plots of posterior samples (gray), the posterior mean (black), 95% credible band (dashed), the Grenander estimator (blue), and $f^\star$ (red).

# Monotone density, cont.

| | | Coverage Prob. | | Mean Length | |
|---|---|---|---|---|---|
| $n$ | $x$ | EB | DPM | EB | DPM |
| 100 | 0.5 | 0.943 | 0.987 | 0.371 | 0.459 |
| | 1.0 | 0.984 | 0.965 | 0.270 | 0.338 |
| | 2.0 | 0.986 | 0.938 | 0.150 | 0.185 |
| | 3.0 | 0.990 | 0.891 | 0.086 | 0.098 |
| 200 | 0.5 | 0.934 | 0.981 | 0.285 | 0.371 |
| | 1.0 | 0.962 | 0.955 | 0.219 | 0.285 |
| | 2.0 | 0.983 | 0.914 | 0.120 | 0.154 |
| | 3.0 | 0.970 | 0.842 | 0.068 | 0.082 |

Table 1: Exponential example—coverage probability and mean length of the 95% posterior credible regions for the empirical Bayes (EB) and Dirichlet process mixture model (DPM), for several values of $x$ and $n$.

# Conclusion

- For high-dim problems, incorporating structure is key.
- We often have relevant information about the structure.
- But a "good" prior for the structure-specific parameters can be difficult to come by.
- Empirical priors have some nice features:
  - good rates and model selection properties
  - promising in uncertainty quantification
  - fast to compute?!?!
- Some general theory available, but more work to do.
- Interesting applications I'm looking at now:
  - sparse precision matrix estimation;
  - general mixture models;
  - high-dim GLMs;
  - ...

# The end... except for a commercial

*Thank you!*

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